

Current Numerical Results: Tables Only

13 March 2026

1. Paper Benchmark: Heston Bermudan Put

Table 1: Benchmark setup for the paper-style Heston case.

Quantity	Value
Model	Heston
Option	Bermudan put
S_0	10
K	10
T	1
r	0.02
v_0	0.15
κ	5
θ	0.16
η	0.9
ρ	0.1
Exercise dates	12
LSMC paths	500,000
LSMC steps M	1,000
Hybrid paths N	10,000
Hybrid low paths N_{low}	10,000
Asset grid resolution	512
Finite-difference reference	1.4507

Table 2: Benchmark results for the paper-style Heston case.

Estimator	Published value	Current run	Run – FD	Run – published
LSMC direct	1.4494	1.458896	0.008196	+0.009496
LSMC low	1.4487	1.452544	0.001844	+0.003844
Hybrid direct	1.4530	1.452988	0.002288	-0.000012
Hybrid low	1.4529	1.445493	0.005207	-0.007407

Table 3: Setup for the current gDMR baseline run.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.50
δ_2	0.50
Exercise dates	100
Euler steps M	600
LSMC paths	1,000,000
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Asset grid points N_S	181
Hermite nodes	64

2. gDMR Baseline: $\delta_1 = \delta_2 = 0.50$

3. gDMR Equal-Delta Case: $\delta_1 = \delta_2 = 0.75$

4. Comparison of the Two gDMR Cases

5. Equal-Delta Strike Tests

Table 4: Results for the current gDMR baseline run.

Estimator	Price	Standard error	Relative error
LSMC direct	6.384535	0.007943	reference
LSMC low	6.387249	0.007939	0.04% vs LSMC direct
Hybrid direct	6.421758	0.000652	0.58% vs LSMC direct
Hybrid low	6.265070	0.045711	1.87% vs LSMC direct

Table 5: Setup for the equal-delta gDMR run.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.75
δ_2	0.75
Exercise dates	100
Euler steps M	600
LSMC paths	1,000,000
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Asset grid points N_S	181
Hermite nodes	64

Table 6: Results for the equal-delta gDMR run.

Estimator	Price	Standard error	Relative error
LSMC direct	6.653560	0.007822	reference
LSMC low	6.655062	0.007819	0.02% vs LSMC direct
Hybrid direct	6.696152	0.000375	0.64% vs LSMC direct
Hybrid low	6.545257	0.045080	1.63% vs LSMC direct

Table 7: Additional comparison metrics for the equal-delta gDMR run.

Quantity	Value
Hybrid low relative error vs LSMC low	1.65%
1% direct target met	yes

Table 8: Comparison between the two gDMR parameter cases.

Estimator	$\delta = 0.50$	$\delta = 0.75$	Change
LSMC direct	6.384535	6.653560	+0.269025
LSMC low	6.387249	6.655062	+0.267813
Hybrid direct	6.421758	6.696152	+0.274394
Hybrid low	6.265070	6.545257	+0.280187

Table 9: Strike-test setup for the equal-delta gDMR runs.

Case	S_0	K
ITM put	100	110
ATM put	100	100
OTM put	100	90

Table 10: Strike-test results for the equal-delta gDMR runs.

Case	Strike	LSMC direct	LSMC low	Hybrid direct	Hybrid low	Hybrid direct rel. error	Hybrid low rel. vs LSMC direct	1% target met
ITM put	110	12.735850	12.734082	12.767870	12.634384	0.25%	0.80%	yes
ATM put	100	6.653560	6.655062	6.696152	6.545257	0.64%	1.63%	yes
OTM put	90	2.743578	2.734472	2.791426	2.668242	1.74%	2.75%	no

Table 11: Additional relative errors for the equal-delta strike tests.

Case	Hybrid low rel. vs LSMC low	LSMC low rel. vs LSMC direct
ITM put	0.78%	0.01%
ATM put	1.65%	0.02%
OTM put	2.42%	0.33%